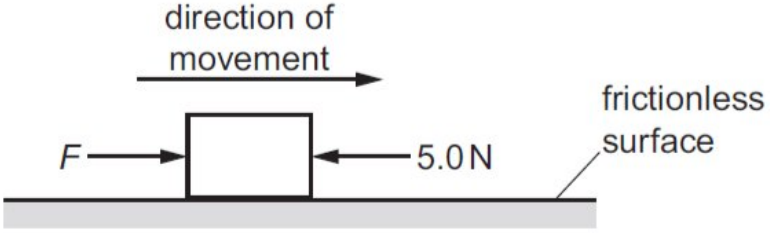
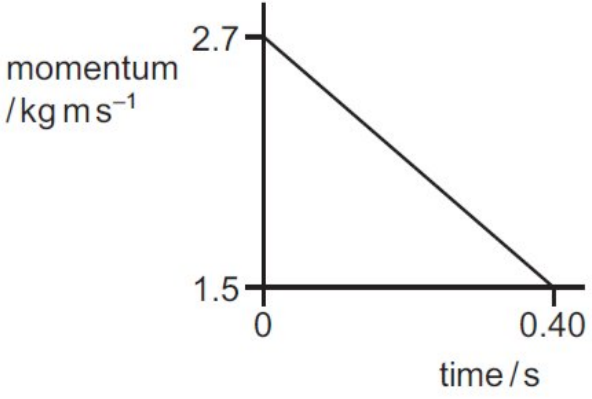
5 A block is moving along a horizontal frictionless surface. A constant force $F$ and a constant resistive force of 5.0 N act on the block as it is moving in the direction of the force $F$, as shown.




The graph shows the variation with time of the momentum of the block.




What is the magnitude of $F$?

**A** 2.0 N **B** 3.0 N **C** 5.0 N **D** 8.0 N